

E1

Consider the circuit in Fig. 1. The circuit parameters at $T = 25^\circ\text{C}$ are the following:

$V_{DD} = 12\text{ V}$; $R_{G1} = 120\text{ k}\Omega$; $R_{G2} = 120\text{ k}\Omega$; $R_D = 100\text{ k}\Omega$; $R_S = 1\text{ k}\Omega$; $R_B = 3\text{ M}\Omega$; $R_F = 33\text{ k}\Omega$; $R_g = 1\text{ k}\Omega$; $C_M = 10\text{ pF}$.

M_1 (e-NMOS): $V_{thN} = 3\text{ V}$; $I_{DSSN} = 0.9\text{ mA}$; $r_0 = +\infty$;

Q_2 (npn BJT): $V_{BE} = 0.7\text{ V}$; $\beta_F = \beta_0 = 200$; $r_0 = +\infty$;

$V_T = 26\text{ mV}$

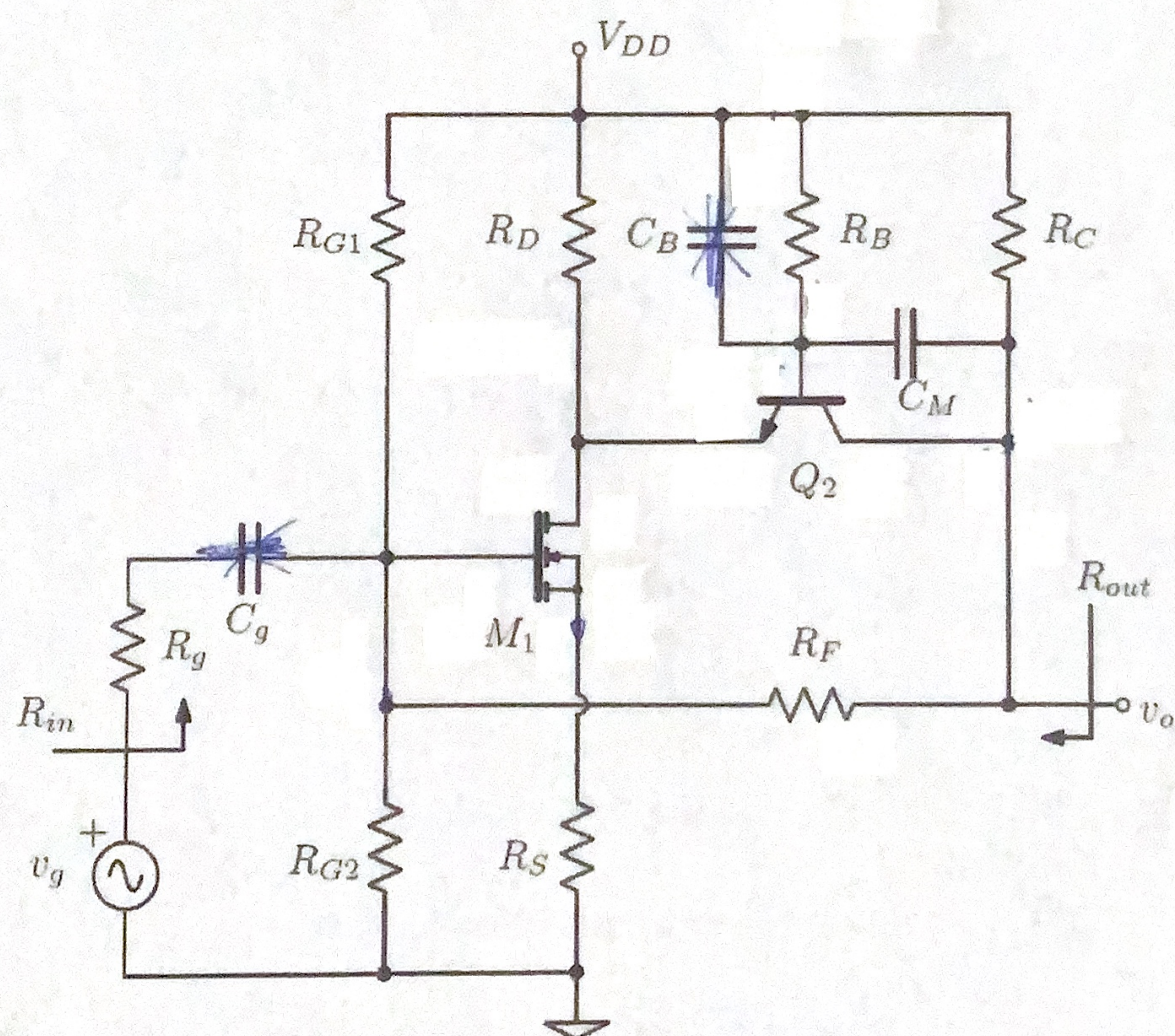


Fig. 1 – A case of two-stage feedback amplifier.

Considering *all capacitors* to be equivalent to *open circuits*, determine:

1. the value of resistance R_C that makes $V_O = V_{DD}/2$ in the steady state;
2. the operating points (V_{DS} , I_D) and (V_{CE} , I_C) of M_1 and Q_2 .

Assuming that, at the frequencies of interest (i.e. at mid-band), C_g and C_B can be considered equivalent to *short circuits*, while C_M can be considered equivalent to an *open circuit*, determine also:

3. the voltage gain $A_v = v_o/v_g$ of the amplifier;
4. the input resistance R_{in} indicated in the figure;

Finally, considering only the effect of capacitor C_M :

5. the high frequency bandwidth limit f_H of the amplifier.

E2

Consider the operational amplifier configuration shown in Fig. 2. The circuit parameters are the following:

$R_1 = 12 \text{ k}\Omega$; $R_2 = 180 \text{ k}\Omega$.

OpAmp differential gain: $A_{d0} = 10^5 \text{ [V/V]}$, $\omega_1 = 100 \text{ rad/s}$, $\omega_2 = 10 \text{ krad/s}$.

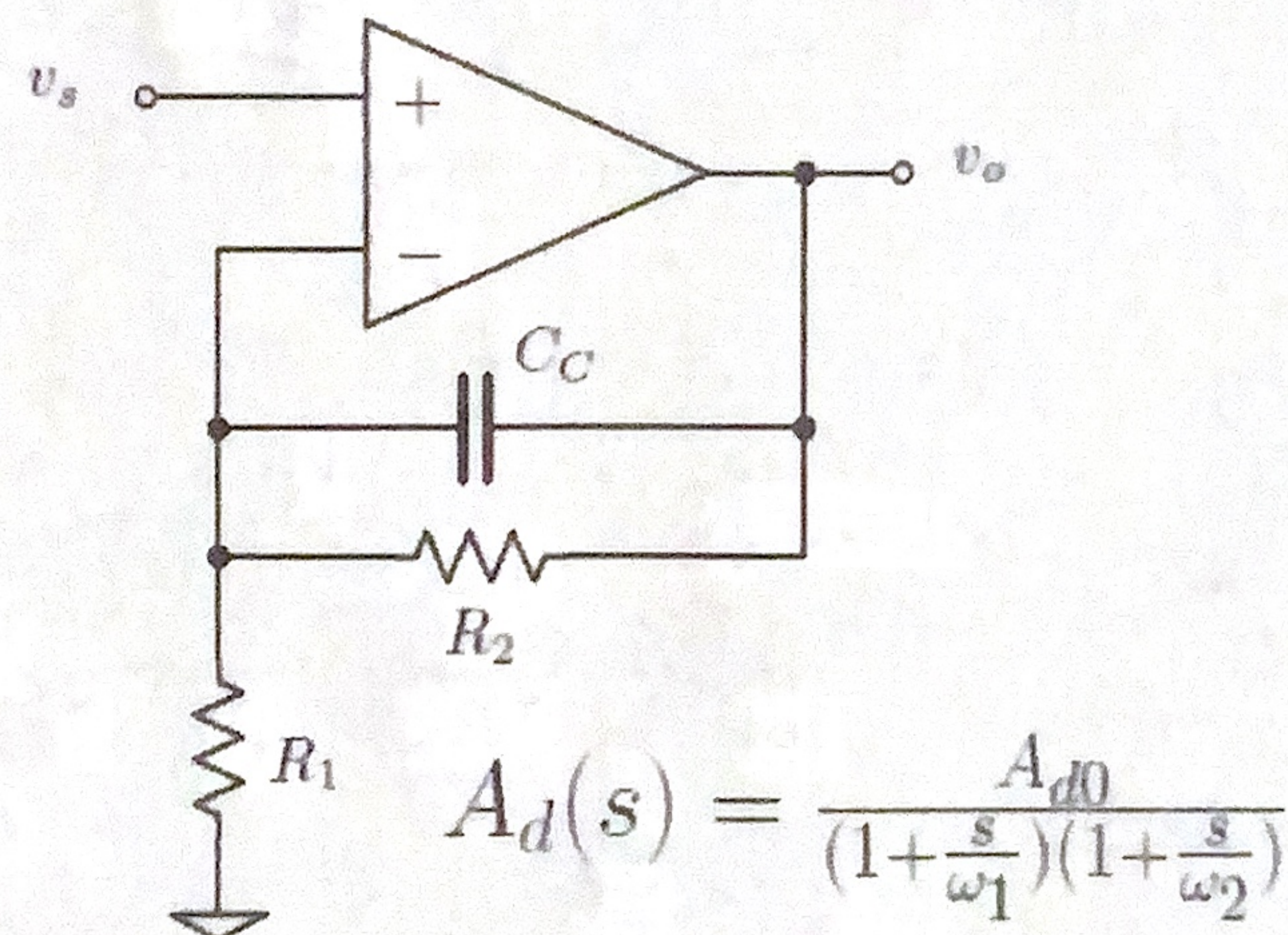


Fig. 2 – Partially compensated OpAmp in the non-inverting configuration.

Considering, initially, C_C **equivalent to an open circuit**, determine:

1. the expression of the *ideal* circuit voltage gain $A_v(s) = v_o/v_s$;
2. the bandwidth and phase margin of the amplifier.

Then, considering the effect of the compensation network, determine

3. the value of C_C needed to get the maximum phase margin boost from the compensation network;
4. the new bandwidth and phase margin of the amplifier.